NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

What Does This Test Support?

AP Statistics · Topic 7.9 (CED 4.10) · B: 2027-01-29 · E: 2027-03-19

[STAR] TODAY'S PROBLEM

Suppose 4,000 active students chose Focus mode and 3,600 chose Guided mode on an online practice service this term. Independent SRSs of 100 nonoverlapping students completed the same set; summaries are shown. Before sampling, the service asked whether $\mu_F > \mu_G$ and set $\alpha = 0.05$. Students chose their modes, and only a difference of at least 3 minutes is practically important.

Nia: "Since $p \approx 0.0263 < 0.05$, there is evidence that the Focus population mean is higher. Therefore Guided mode causes every student to finish at least 3 minutes faster; switch everyone."

Mateo: "The observed gap is only 1.2 minutes and students selected their modes, so neither a 3-minute effect nor causation is established. Therefore retain H_0; there is no evidence of a population mean difference."

Practice time summaries (minutes)

Mode	N	n	$\bar{x}$	s
Focus	4,000	100	42.8	4.2
Guided	3,600	100	41.6	4.5

FIRST TAKE — before the video (5 min)

Which clauses in Nia's and Mateo's reports look defensible? Cite one number and one design feature.

[BOOK] CARRY OUT THE TEST, THEN BOUND THE CONCLUSION (keep on the page)

For independent samples, $SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$ and $t = ((\bar{x}_1 - \bar{x}_2) - 0)/SE$. A p-value is the probability, assuming H_0, of a statistic at least as extreme in the direction of H_a. If $p \leq \alpha$, reject H_0 and state evidence for H_a in context. Significance against 0 does not establish practical size or an individual guarantee. Random sampling supports population scope; random assignment, not self-selection, supports causation.

Finish after the video:

DOK 1 **(a)** Define μ_F and μ_G, then state H_0 and H_a in Focus-minus-Guided order.

DOK 2 **(b)** Check the two-sample t conditions. Compute the difference, SE, t, technology degrees of freedom, and one-sided p-value; interpret the p-value.

DOK 3 **(c)** Evaluate both reports and write one warranted conclusion. Coordinate p versus α, the 1.2-minute gap versus the 3-minute benchmark, means versus individuals, SRS population scope, and self-selection. Explain the decision risk of retaining either report unchanged.

Frame: The test supports ____________________ because ____________________; it does not establish ____________________ because ____________________.

EXIT — turn this in

One thing the video changed about my first take: ______________________________